SUNIX SDC Expansion Board

libgpiod

Sample Code Guideline

Driver Ver. 2.1.1.0

Doc Ver. 2.0

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1). C/C++

1.1 libgpiod Sample

1.1.1Tutorial

About libgpiod C/C++ programming tutorial, please check readthecocs.io libgpiod tutorial:

https://libgpiod.readthedocs.io/en/latest/

1.1.2. Sample Code

1). This document will use SDC 8DIO loopback as an example. The hardware wiring method is as follows.

DIO Connection mode: Loopback
DO PWR ↔ DI COM
DO GND ↔ DI GND
DO1 ↔ DI1
DO2 ↔ DI2
DO3 ↔ DI3
DO4 ↔ DI4
DO5 ↔ DI5
DO6 ↔ DI6
DO7 ↔ DI7
DO8 ↔ DI8

2). Confirm that the following packages have been installed:

- gpiod

Ubuntu/Debian

```
$apt-get install gpiod libgpiod-dev
```

3). Enter the GPIO C sample project directory.

./SDCLinuxExpansionBoardDriver_VX.X.X.X/GPIO/c/gpio_test

4). Compile the source code and execute the program, the main page show the available GPIO chips. Please ensure that there is GPIO for SDC, in this case, there are 16 lines, because the device is 8DI/8DO.

$make

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_[redacted]GPIO/c/gpio_test# ls
gpio_test.c  main.c  Makefile  obj  precomp.h  utils.c
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_[redacted]GPIO/c/gpio_test# make
gcc -Wall -g -c main.c
gcc -Wall -g -c gpio_test.c
gcc -Wall -g -c utils.c
mv -f *.o obj
gcc -g -o gpio_test \
obj/main.o \
obj/gpio_test.o \
obj/utils.o \
-lgpiod -pthread
rm -f obj/*.o
mv -f gpio_test obj
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_V[redacted]GPIO/c/gpio_test# ./obj/gpio_test
```

```
=======================================================
Available GPIO chips in the system:
Name            Label                               Lines
gpiochip0       INTC1085:00                         473
gpiochip1       sdc_gpio.2.auto                     16
=======================================================
>>-------------------------------------<<
     A : GPIO Chip Open
     B : GPIO Chip Close
     C : Get Chip Line List
     D : Set Output Value
     E : Get Input Value
     F : Wait Input Event

     Q : quit
>>-------------------------------------<<


>> CMD ?
```

5). Press "A" open the $gpiochip_n$ that is SDC hardware device. Just input the number n.

```
=======================================================
Available GPIO chips in the system:
Name            Label                               Lines
gpiochip0       INTC1085:00                         473
gpiochip1       sdc_gpio.2.auto                     16
=======================================================
>>-------------------------------------<<
     A : GPIO Chip Open
     B : GPIO Chip Close
     C : Get Chip Line List
     D : Set Output Value
     E : Get Input Value
     F : Wait Input Event

     Q : quit
>>-------------------------------------<<


>> CMD ?

GPIO device number(e.g if /dev/gpiochip1, input 1)                         ? 1
Open /dev/gpiochip1 Successfully
```

6). Press “C” to dump the line list.

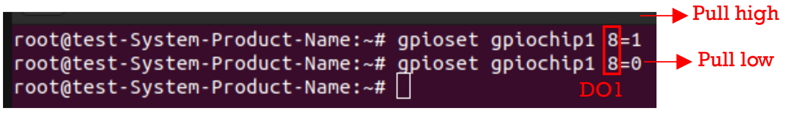
```
>> CMD ?

========== /dev/gpiochip1:16 lines ==========
line0    direction:Input
line1    direction:Input
line2    direction:Input
line3    direction:Input
line4    direction:Input
line5    direction:Input
line6    direction:Input
line7    direction:Input
line8    direction:Output
line9    direction:Output
line10   direction:Output
line11   direction:Output
line12   direction:Output
line13   direction:Output
line14   direction:Output
line15   direction:Output
=============================================
>> CMD ?
```

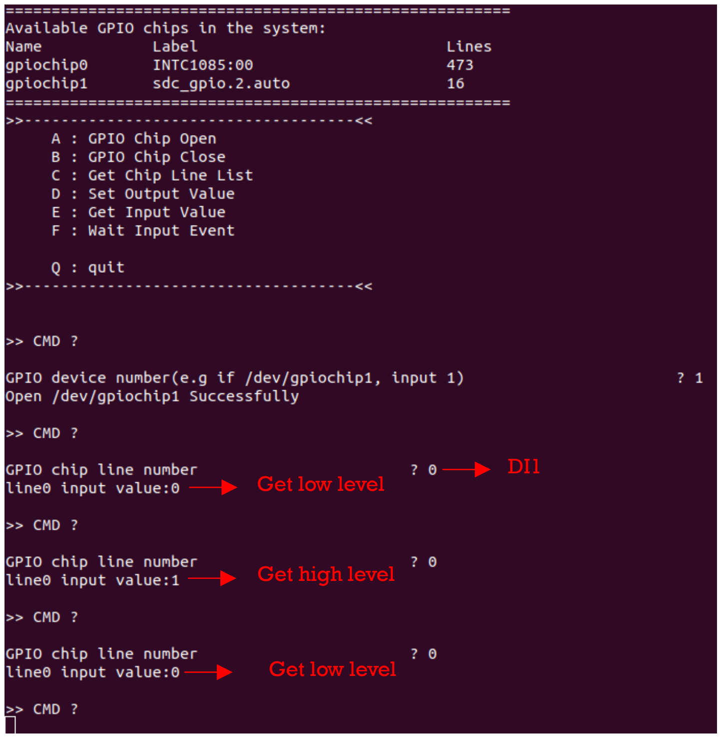
7). Press “D” to set output value.

```
=======================================================
Available GPIO chips in the system:
Name            Label                           Lines
gpiochip0       INTC1085:00                     473
gpiochip1       sdc_gpio.2.auto                 16
=======================================================
>>-----------------------------------<<
     A : GPIO Chip Open
     B : GPIO Chip Close
     C : Get Chip Line List
     D : Set Output Value
     E : Get Input Value
     F : Wait Input Event

     Q : quit
>>-----------------------------------<<


>> CMD ?

GPIO device number(e.g if /dev/gpiochip1, input 1)          ? 1
Open /dev/gpiochip1 Successfully

>> CMD ?

GPIO chip line number                       ? 8  → DO1
Output value(0 or 1)                        ? 1  → Pull high
Set output value successfully, line:8, GPIO chip:/dev/gpiochip1

>> CMD ?

GPIO chip line number                       ? 8
Output value(0 or 1)                        ? 0 → Pull low
Set output value successfully, line:8, GPIO chip:/dev/gpiochip1

>> CMD ?
```

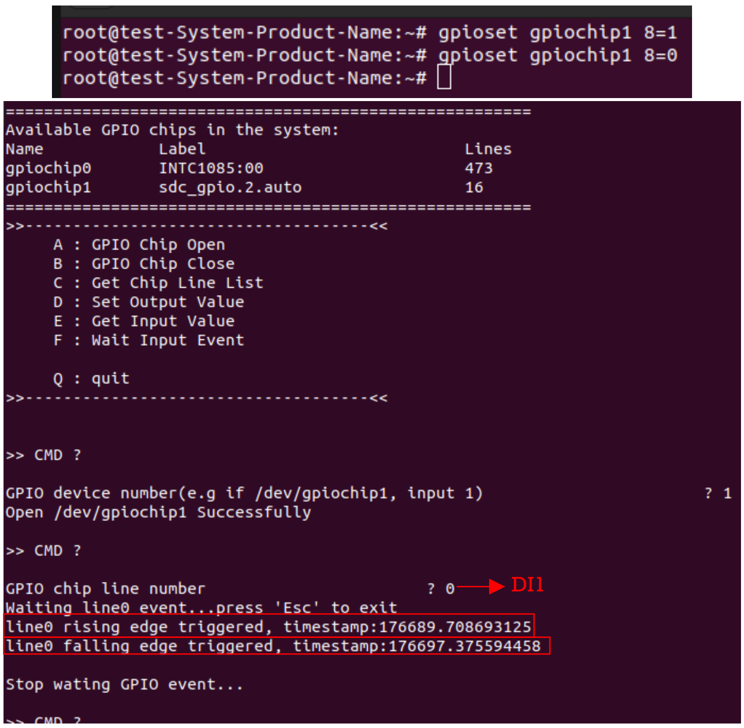
```
root@test-System-Product-Name:~# gpioget gpiochip1 0
0 → Get low level                                  DI1
root@test-System-Product-Name:~# gpioget gpiochip1 0
1 → Get high level
root@test-System-Product-Name:~# gpioget gpiochip1 0
0 → Get low level
root@test-System-Product-Name:~#
```

8). Press “E” to get input value.





9). Press “F” to waiting the input event, after test finish, press ‘Esc’ key to stop waiting.



2). Python

2.1 gpiod Module

2.1.1 Tutorial

About libgpiod python programming tutorial, please check pypi.org and Github tutorial.

https://pypi.org/project/gpiod/

https://github.com/brgl/libgpiod/tree/master/bindings/python/examples

2.1.2 Sample Code

1). Install python3.

```
$apt-get install python3
```

2). Install pip.

```
$apt-get install python3-pip
```

3). Install python-can module by pip.

```
$pip3 install gpiod
```

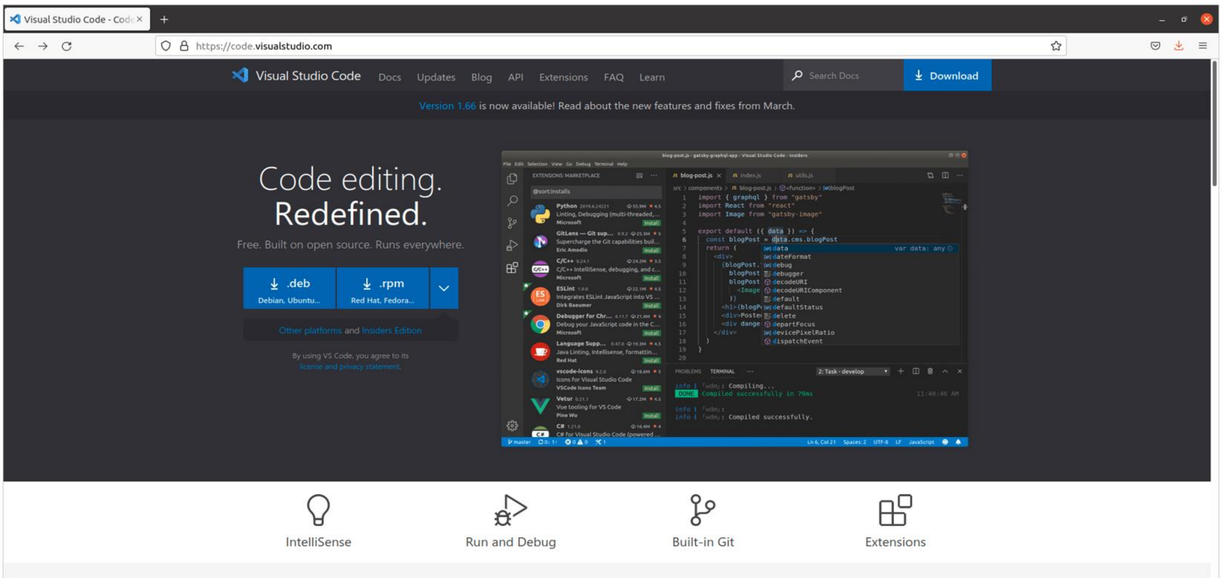
4). Install keyboard module by pip

```
$pip3 install keyboard
```

5). Install click module by pip

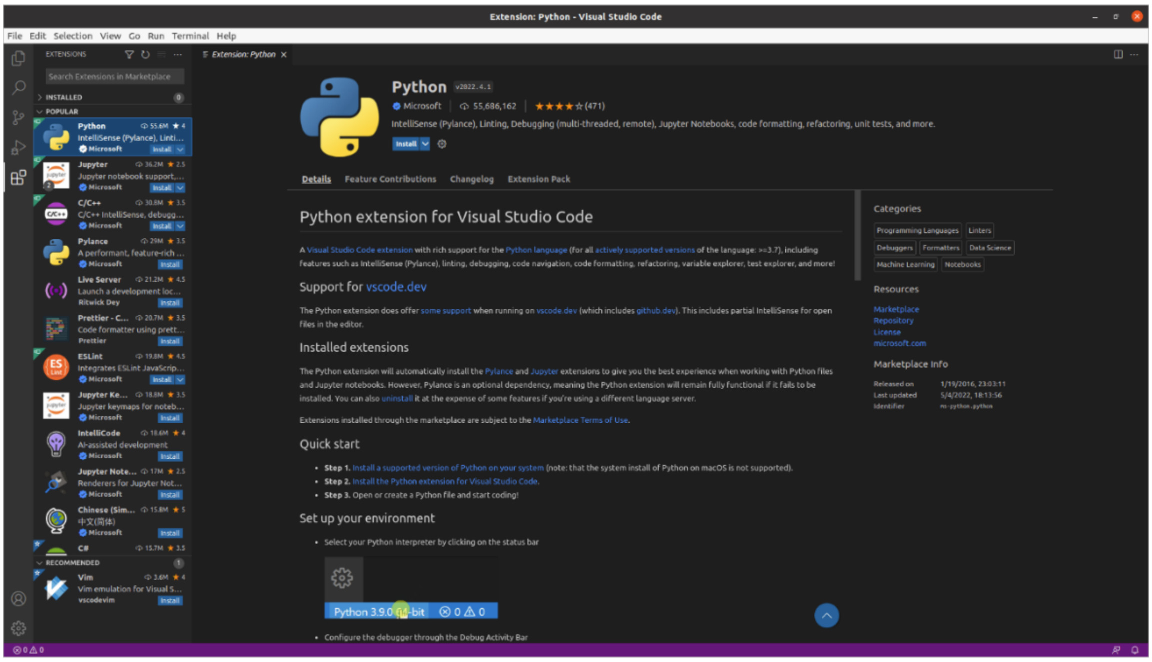
```
$pip3 install click
```

6). Install IDE, This case use Visual Studio Code as example, developer could download the deb file by official website and install it.



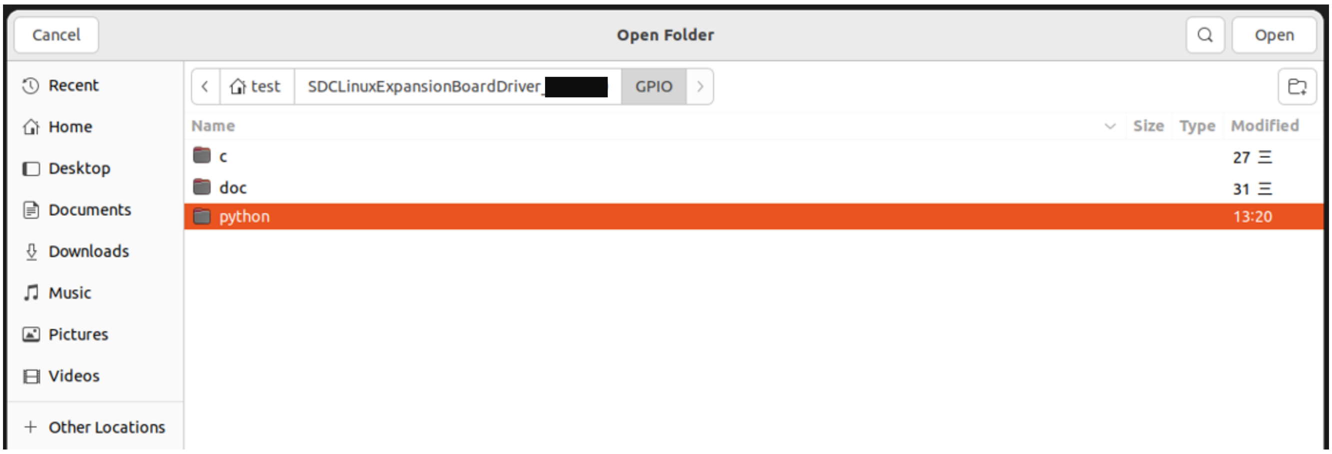
Anyway, developer can choose IDE according to your preferences, e.g. PyCharm, Spyder, Atom, Eclipse, or even vim.

7). Install the Python development kit for VScode.

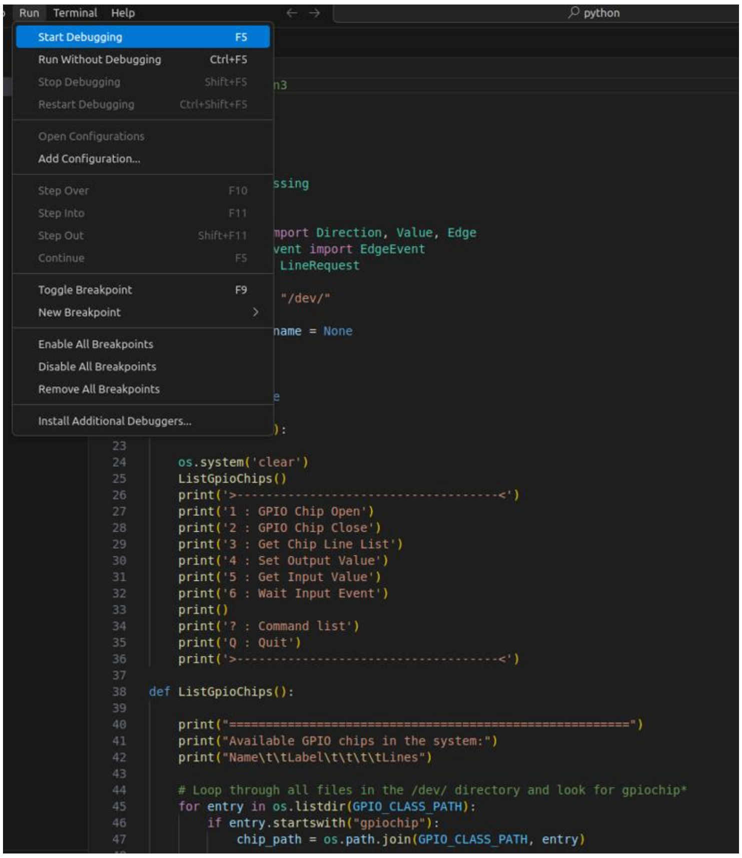


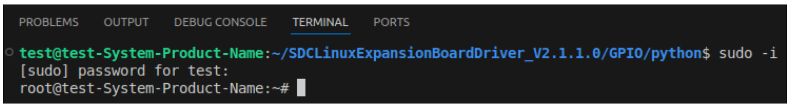
8). Open the folder which content the gpio_test.py sample
The directory:

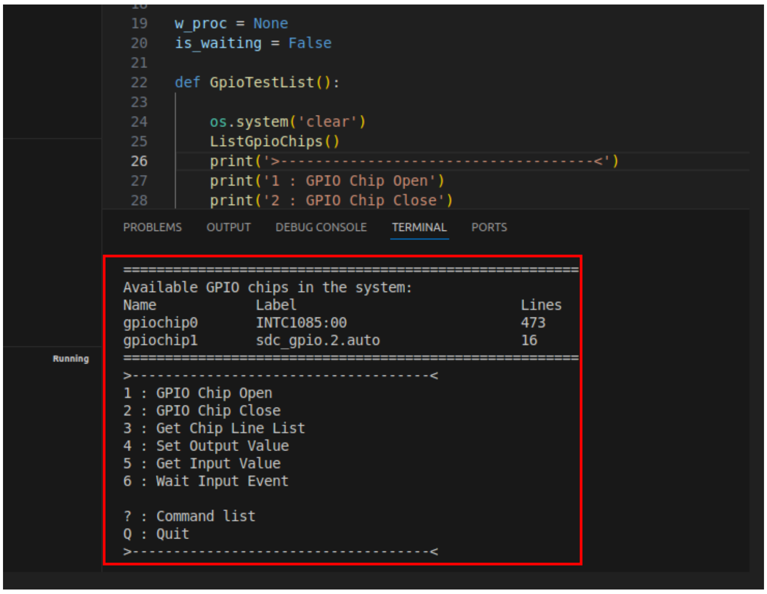
./ SDCLinuxExpansionBoardDriver_VX.X.X.X /GPIO/python/gpio_test.py



9). Start to debug gpio_test.py, must be run as root.







10). Stop VSCode debug, run gpio_test.py on terminal. he main page show the available GPIO chips. Please ensure that there is GPIO for SDC, in this case, there are 16 lines, because the device is 8DI/8DO.

```
$./gpio_test.py
```

```
root@test-System-Product-Name: /home/test/SDCLinuxExpansionBoardDriver_V2.1.1.0/GPIO/python
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_[redacted]/GPIO/python# ls
gpio_test.py
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_[redacted]/GPIO/python# ./gpio_test.py
```

```
=======================================================
Available GPIO chips in the system:
Name            Label                            Lines
gpiochip0       INTC1085:00                      473
gpiochip1       sdc_gpio.2.auto                  16
=======================================================
>-------------------------------------<
1 : GPIO Chip Open
2 : GPIO Chip Close
3 : Get Chip Line List
4 : Set Output Value
5 : Get Input Value
6 : Wait Input Event

? : Command list
Q : Quit
>-------------------------------------<

CMD?
```

11). Press "1" open the $gpiochip_n$ that is SDC hardware device. Just input the number *n*.

```
=======================================================
Available GPIO chips in the system:
Name            Label                            Lines
gpiochip0       INTC1085:00                      473
gpiochip1       sdc_gpio.2.auto                  16
=======================================================
>-------------------------------------<
1 : GPIO Chip Open
2 : GPIO Chip Close
3 : Get Chip Line List
4 : Set Output Value
5 : Get Input Value
6 : Wait Input Event

? : Command list
Q : Quit
>-------------------------------------<

CMD?
GPIO device number(e.g if /dev/gpiochip1, input 1)?
1
Open /dev/gpiochip1 Successfully

CMD?
```

12). Press “3” to dump the line list.

```
CMD?
===========/dev/gpiochip1 lines:16===========
line0    direction: INPUT
line1    direction: INPUT
line2    direction: INPUT
line3    direction: INPUT
line4    direction: INPUT
line5    direction: INPUT
line6    direction: INPUT
line7    direction: INPUT
line8    direction: OUTPUT
line9    direction: OUTPUT
line10   direction: OUTPUT
line11   direction: OUTPUT
line12   direction: OUTPUT
line13   direction: OUTPUT
line14   direction: OUTPUT
line15   direction: OUTPUT
============================================
```

13). Press “4” to set output value.

```
=======================================================
Available GPIO chips in the system:
Name            Label                          Lines
gpiochip0       INTC1085:00                    473
gpiochip1       sdc_gpio.2.auto                16
=======================================================
>------------------------------------<
1 : GPIO Chip Open
2 : GPIO Chip Close
3 : Get Chip Line List
4 : Set Output Value
5 : Get Input Value
6 : Wait Input Event

? : Command list
Q : Quit
>------------------------------------<

CMD?
GPIO device number(e.g if /dev/gpiochip1, input 1)?
1
Open /dev/gpiochip1 Successfully

CMD?
GPIO chip line number  ?
8 ──▶ DO1
Output value(0 or 1)?
1 ──▶ Pull high
/dev/gpiochip1 line8 set value:1

CMD?
GPIO chip line number  ?
8
Output value(0 or 1)?
0 ──▶ Pull low
/dev/gpiochip1 line8 set value:0

CMD?
```

```
root@test-System-Product-Name:~# gpioget gpiochip1 0
0 ──▶ Get low level                              DI1
root@test-System-Product-Name:~# gpioget gpiochip1 0
1 ──▶ Get high level
root@test-System-Product-Name:~# gpioget gpiochip1 0
0 ──▶ Get low level
root@test-System-Product-Name:~#
```

14). Press “5” to get input value.

```
root@test-System-Product-Name:~# gpioset gpiochip1 8=1  ──▶ Pull high
root@test-System-Product-Name:~# gpioset gpiochip1 8=0  ──▶ Pull low
root@test-System-Product-Name:~#                  DO1
```

```
======================================================
Available GPIO chips in the system:
Name            Label                           Lines
gpiochip0       INTC1085:00                     473
gpiochip1       sdc_gpio.2.auto                 16
======================================================
>------------------------------------<
1 : GPIO Chip Open
2 : GPIO Chip Close
3 : Get Chip Line List
4 : Set Output Value
5 : Get Input Value
6 : Wait Input Event

? : Command list
Q : Quit
>------------------------------------<

CMD?
GPIO device number(e.g if /dev/gpiochip1, input 1)?
1
Open /dev/gpiochip1 Successfully

CMD?
GPIO chip line number  ?
0 ──▶ DI1
/dev/gpiochip1 line0 get value:0 ──▶ Get low level

CMD?
GPIO chip line number  ?
0
/dev/gpiochip1 line0 get value:1 ──▶ Get high level

CMD?
GPIO chip line number  ?
0
/dev/gpiochip1 line0 get value:0 ──▶ Get low level

CMD?
```

15). Press “6” to waiting the input event, after test finish, press ‘Esc’ key to stop waiting.

```
root@test-System-Product-Name:~# gpioset gpiochip1 8=1
root@test-System-Product-Name:~# gpioset gpiochip1 8=0
root@test-System-Product-Name:~#
```

```
======================================================
Available GPIO chips in the system:
Name            Label                           Lines
gpiochip0       INTC1085:00                     473
gpiochip1       sdc_gpio.2.auto                 16
======================================================
>------------------------------------<
1 : GPIO Chip Open
2 : GPIO Chip Close
3 : Get Chip Line List
4 : Set Output Value
5 : Get Input Value
6 : Wait Input Event

? : Command list
Q : Quit
>------------------------------------<

CMD?
GPIO device number(e.g if /dev/gpiochip1, input 1)?
1
Open /dev/gpiochip1 Successfully

CMD?
GPIO chip line number  ?
0 ──▶ DI1
Waiting line0 event...press "esc" key to exit.
line0 rising edge triggered. Offset:0, Timestamp:99.163484479
line0 falling edge triggered. Offset:0, Timestamp101.755450169
^[
CMD?
```